

Disproof of Conjecture 00000000591

(New direction on Wilf's conjecture for embedding dimension 3)

TLMC submission

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1 The conjecture and the verdict

Wilf's conjecture states $e \cdot n(S) \geq g(S) + e$ for a numerical semigroup S with embedding dimension e , Frobenius number $g(S)$ and $n(S) = \#\{s \in S : 0 \leq s \leq g(S)\}$. Conjecture 00000000591 adds a *new direction*: on embedding-dimension-3 semigroups the *surplus* satisfies

$$\frac{3n(S) - g(S) - 3}{g(S)} \xrightarrow{g(S) \rightarrow \infty} 0 \quad \text{with an explicit convergence rate.}$$

Verdict: FALSE. The ratio stays near $1/2$ along infinite families of embedding-dimension-3 semigroups with $g \rightarrow \infty$; it is in fact universally bounded above by $(g-3)/(2g)$, and this ceiling is attained exactly by an explicit family. Note Wilf's inequality itself holds with room to spare on every example below; it is the vanishing-rate claim that fails.

2 A universal ceiling, and a family that attains it

Claim 1 (pairing bound). *Let S be any numerical semigroup with Frobenius number g . For $0 < x < g$, x and $g-x$ cannot both lie in S , else $g = x + (g-x) \in S$, a contradiction. If g is even then $g/2 \notin S$. Hence $n(S) \leq \lceil g/2 \rceil$ and*

$$3n(S) - g - 3 \leq \frac{3(g+1)}{2} - g - 3 = \frac{g-3}{2}, \quad \text{i.e.} \quad \frac{3n(S) - g - 3}{g} \leq \frac{g-3}{2g} < \frac{1}{2}.$$

Theorem 1 (counterexample family). *Let p, q, r be distinct primes and $S = \langle pq, pr, qr \rangle$. Then S has embedding dimension 3, is a complete intersection (hence symmetric), and*

$$g(S) = 2pqr - pq - pr - qr, \quad n(S) = \frac{g(S) + 1}{2}, \quad 3n(S) - g(S) - 3 = \frac{g(S) - 3}{2}.$$

Therefore $\frac{3n(S) - g(S) - 3}{g(S)} = \frac{g(S) - 3}{2g(S)} \rightarrow \frac{1}{2} \neq 0$ as $pqr \rightarrow \infty$. The convergence-to-0 claim fails maximally.

Both formulas were verified computationally on every prime triple with $200 \leq g \leq 5000$ (see `reproduce.py`, Section 2), including the exact Frobenius number $g = 2pqr - pq - pr - qr$ in each case.

3 Computed instances (independently recomputed)

S	g	n	surplus	$3n - g - 3$	surplus/ g
$\langle 17, 23, 29 \rangle$	215	104		94	0.4372
$\langle 9, 13, 101 \rangle$	95	48		46	0.4842
$\langle 15, 33, 55 \rangle$	227	114		112	0.4934
$\langle 15, 39, 65 \rangle$	271	136		134	0.4945
$\langle 35, 55, 77 \rangle$	603	302		300	0.4975
$\langle 401, 403, 405 \rangle$	80 599	40 200		120 598	0.4963
$\langle 10403, 10807, 11021 \rangle$	2 194 011	1 097 006		1 097 004	0.4999993

The last row is the symmetric family at $p, q, r = 101, 103, 107$: $g \approx 2.2 \times 10^6$ and the ratio equals $(g - 3)/(2g)$ to seven decimals. A sweep of generic triples $6 \leq a < b < c \leq 30$ with $g \geq 200$ gives 97 admissible semigroups, 93 of which have surplus/ $g \in [0.40, 0.505]$ — the ratio is statistically pinned near $1/2$, never near 0.

4 Lean 4 verification

Directory `lean4/` (core Lean 4.33.1, no Mathlib). All proofs are pure kernel computations (`rfl/decide`); `Check.lean` audits 21 theorems with `#print axioms` and all report “*does not depend on any axioms*”; there is no `sorry`. Verified facts, one theorem group per semigroup (a, b, c) :

- `rep a b c g = false`: g is a gap (membership `rep` is an exhaustive bounded search over $y \leq n/b, z \leq n/c$, remainder divisible by a);
- `cover`: $\min(a, b, c)$ consecutive integers above g are nongaps (with closure under addition this certifies g is *the* Frobenius number);
- `count`: the nongap count n on $[0, g]$ equals the stated value;
- `surplus`: $3n - g - 3$ equals the stated value;
- `ratio`: $2g < 5 \cdot \text{surplus}$, i.e. $\text{surplus}/g > 2/5 > 0$.

The combined statement `main` conjoins all 20 facts for $\langle 17, 23, 29 \rangle$, $\langle 9, 13, 101 \rangle$, $\langle 15, 33, 55 \rangle$, $\langle 35, 55, 77 \rangle$.

5 Reproducing

```
python3 reproduce.py
cd lean4
lake build
lake env lean Check.lean
```